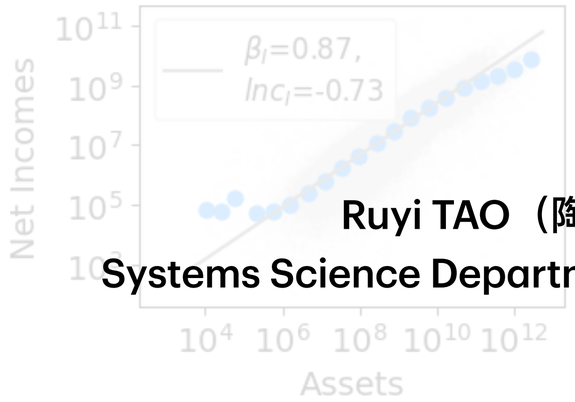
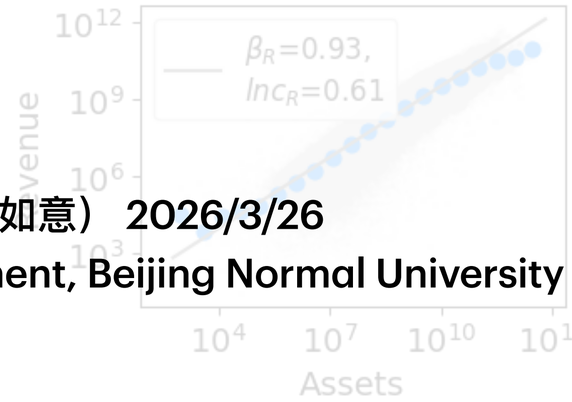
# Company Growth Modeling Based on Scaling Theory










Ruyi TAO（陶如意） 2026/3/26
Systems Science Department, Beijing Normal University

# 1. Background

◀ Companies are the basic units of economic activities



Exter-Interaction network
(Trade, finance, etc.)




**Company Growth Theory?**

Increasing the size and
complexity




Inter-activities
(Accounting, value added, etc.)

heterogeneity

Chaotic

nonlinearity

Random

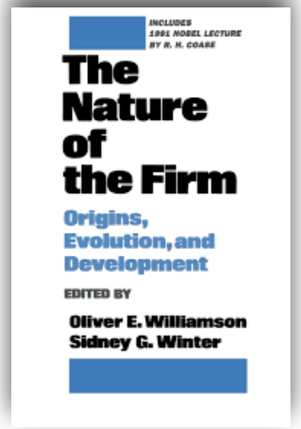
# 2. Current Progress

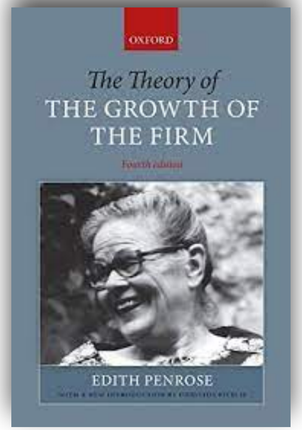
## Company growth theory

The issue of “company growth” has a long history. Questions such as "why do companies grow" and "how to guide company growth" have long been the focus in the fields of economics and management.

Economic models rely on strong assumptions, emphasize theory, and are difficult to test.

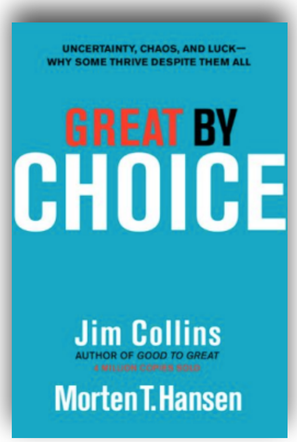
Management focuses on individual cases rather than "laws".









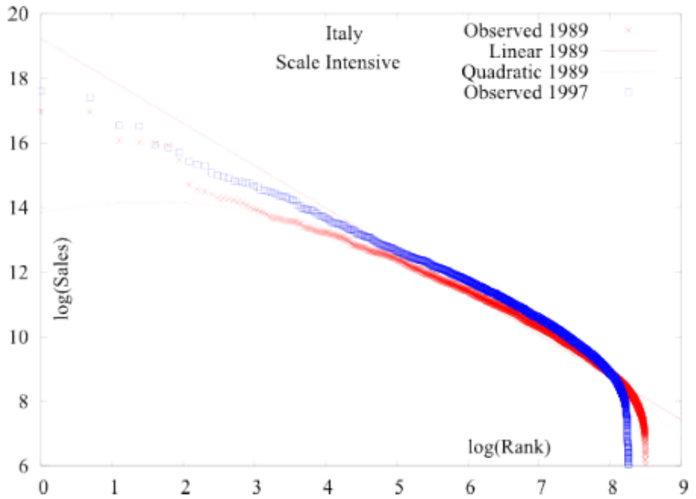

A lack of 「scientific」 models (with reasonable, verifiable assumptions and results)

# 2. Current Progress

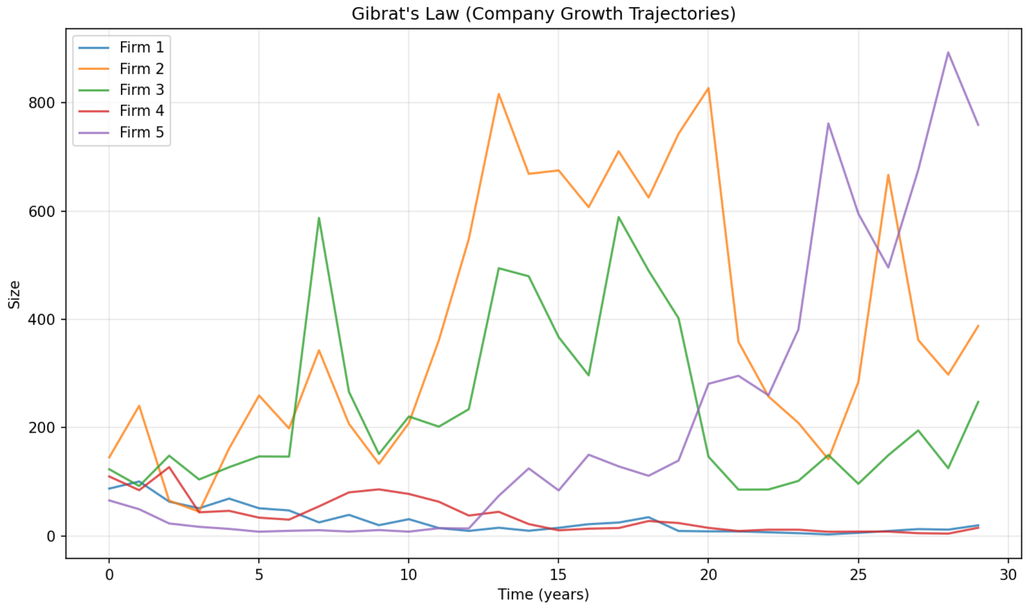
## Quantitative growth model

Gibrat's law[1937] and its derivative models[Sutton, J. review, 1997.]

$$dX_t = \mu(X_t, t)\,dt + \sigma(X_t, t)\,dz_t$$




- The skewed distribution of size
- The lognormal distribution of the growth rate






# 2. Current Progress

## Quantitative growth model

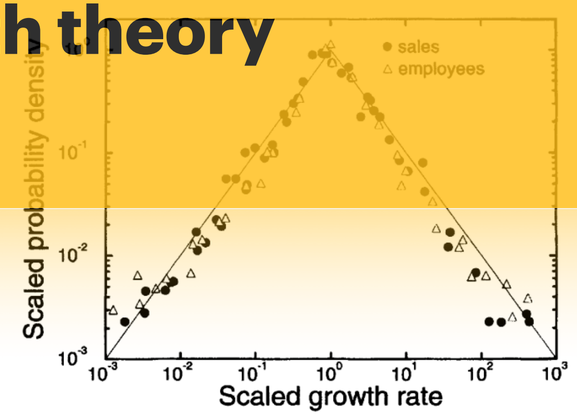
Econophysics model

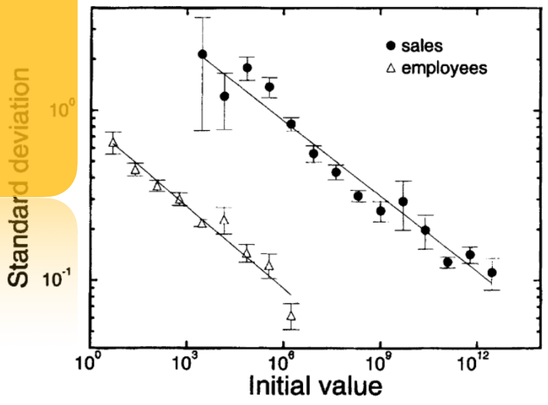
Fu, D.,et al. 2005.
Gabaix, X. 2011.
Moran, J., et al.2024




- The tend-shape distribution of the growth rate
- The power-law between growth rate fluctuations and size

**We still don't have a Predictable, Verifiable and Explainable company growth theory**

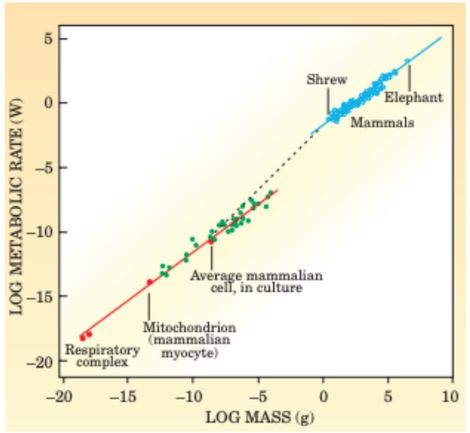


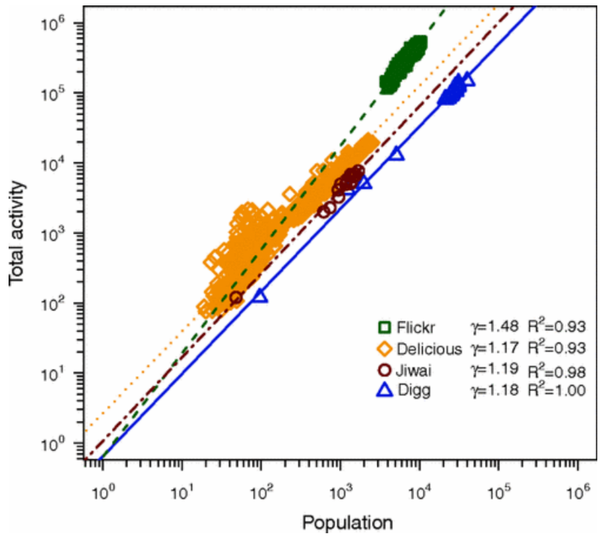




Pay more attention to statistical behaviors, such as distribution of size, growth rate, fluctuations of growth rate, etc. However, they are very coarse-grained description and cannot predict or guide the growth of company

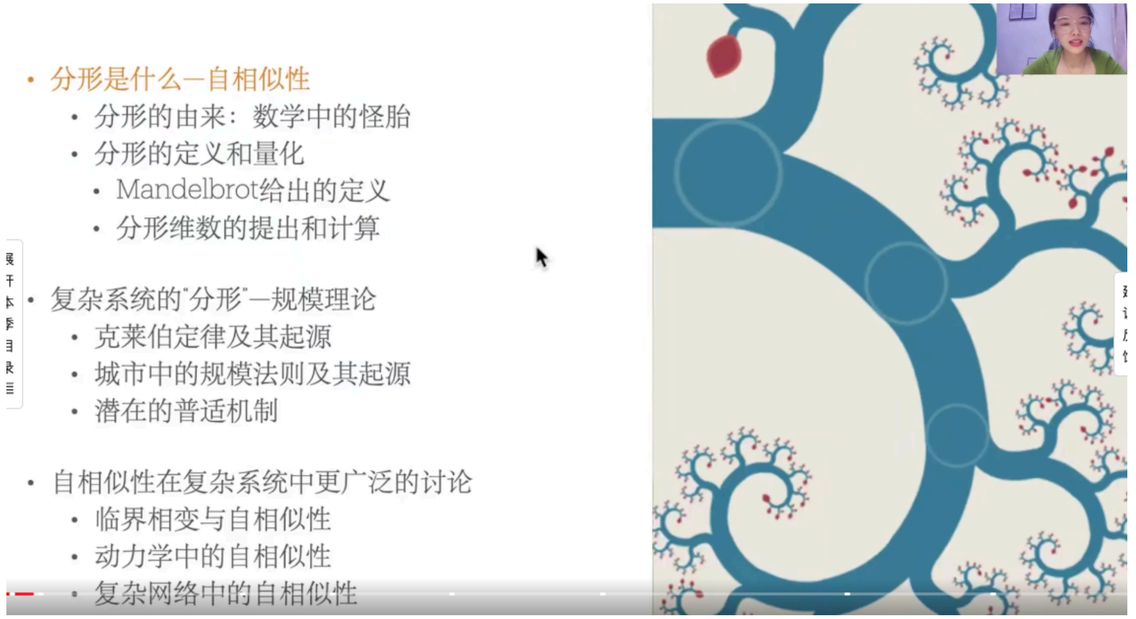
# Company Growth Modeling Based on Scaling Theory

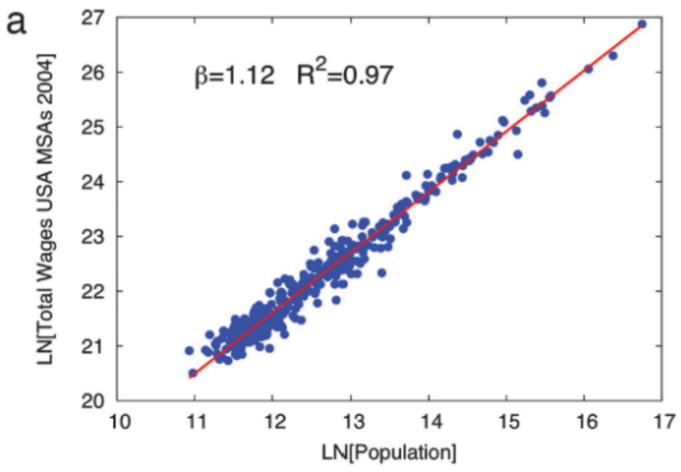
## What is Scaling law

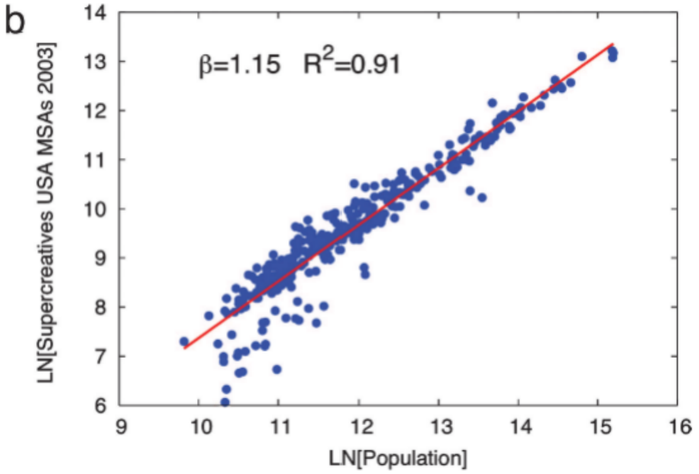



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$$Y = cX^{\beta}$$

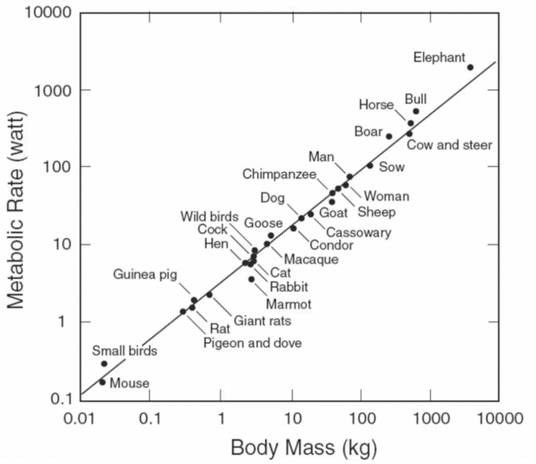
- Scale invariance (fractal)
- Complexity

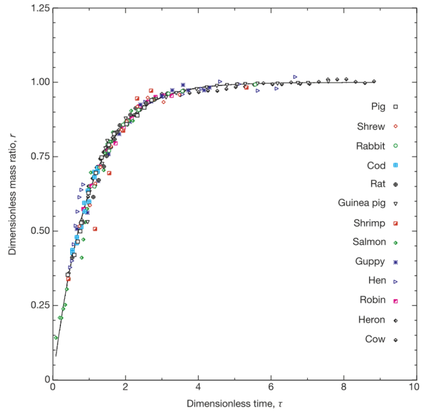
# Company Growth Modeling Based on Scaling Theory

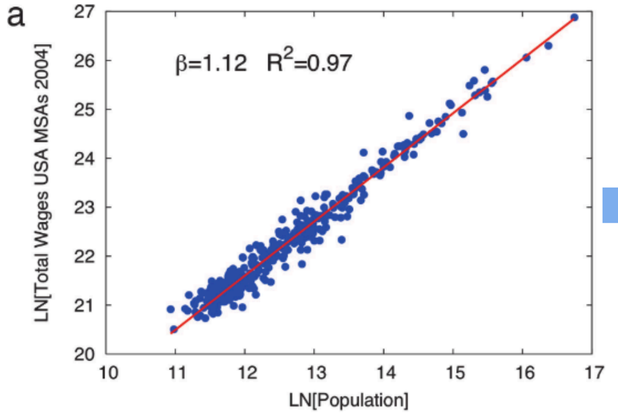
## What is Scaling Theory

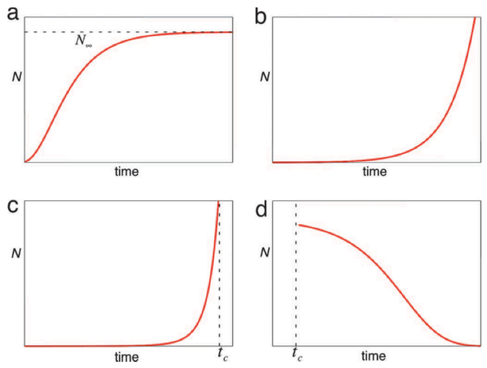
From Kleiber's Law of Biological Systems to Biological Growth.
West,Nature, 2001

From scaling law in city to the city growth equation
Bettencourt, PNAS, 2007



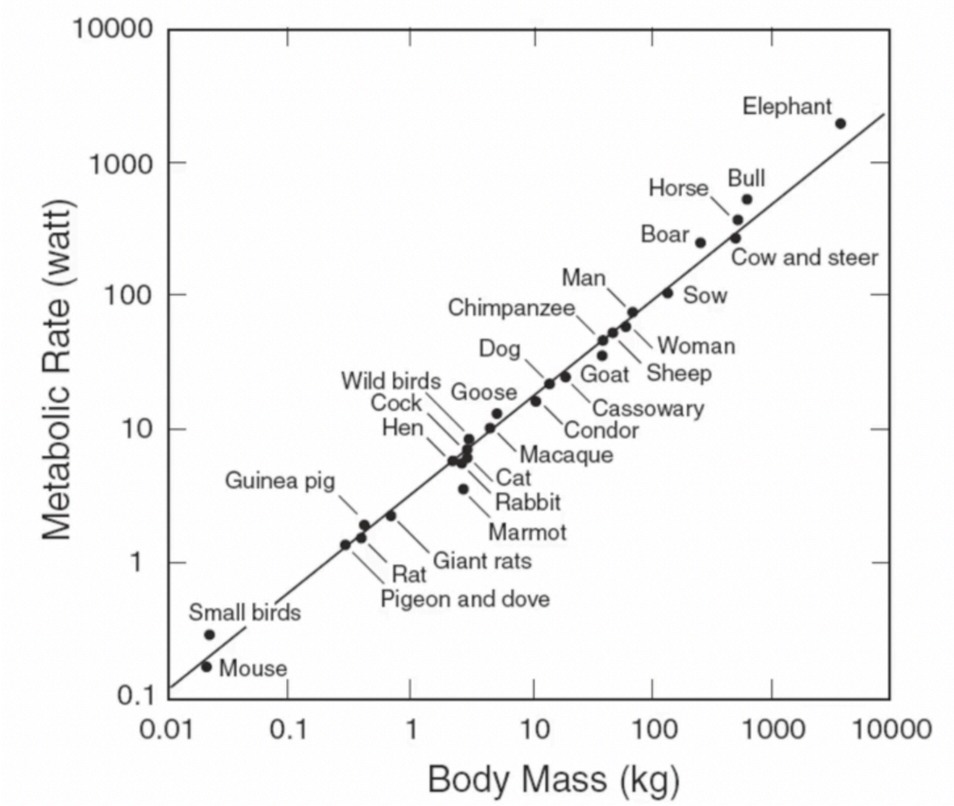




Scaling Theory: Built the bridge of scaling law to system growth

# Company Growth Modeling Based on Scaling Theory

## Scaling theory in biology




Kleiber's law: the **metabolic rate** of an organism is related to its **biomass** to the power of 3/4

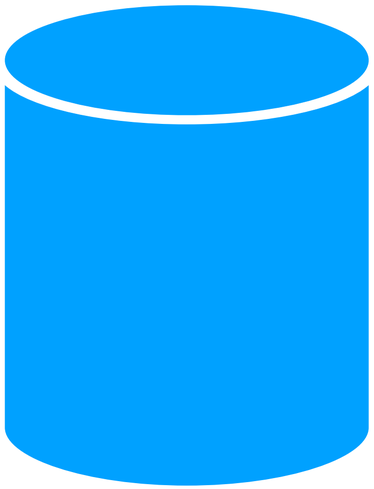
$$B = cM^{3/4}$$

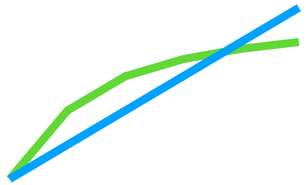
West, G. B., Brown, J. H., & Enquist, B. J. (1997). A general model for the origin of allometric scaling laws in biology. Science, 276(5309), 122–126. https://doi.org/10.1126/science.276.5309.122

# Company Growth Modeling Based on Scaling Theory

## Scaling theory in biology

$$B \propto M^{3/4}$$

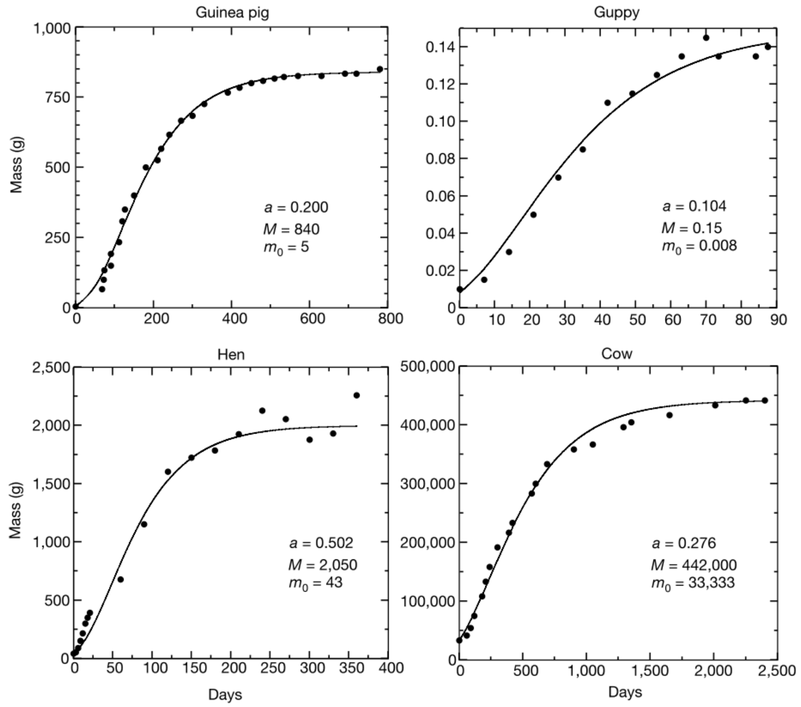


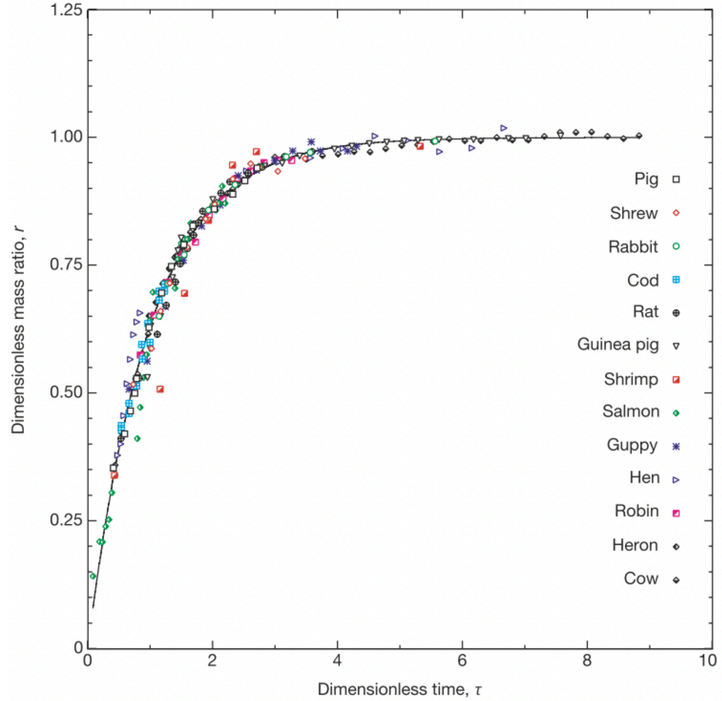


$$B \propto M$$

$$\frac{dm}{dt} = \left(\frac{m_c}{E_c}\right)B - \left(\frac{B_c}{E_c}\right)m$$

$$\left(\frac{m}{M}\right)^{1/4} = 1 - \left[1 - \left(\frac{m_0}{M}\right)^{1/4}\right]e^{-at/4M^{1/4}}$$




West, G. B., Brown, J. H., & Enquist, B. J. (2001). A general model for ontogenetic growth. *Nature*, *413*(6856), 628–631. https://doi.org/10.1038/35098076

# Company Growth Modeling Based on Scaling Theory

## Scaling theory in City

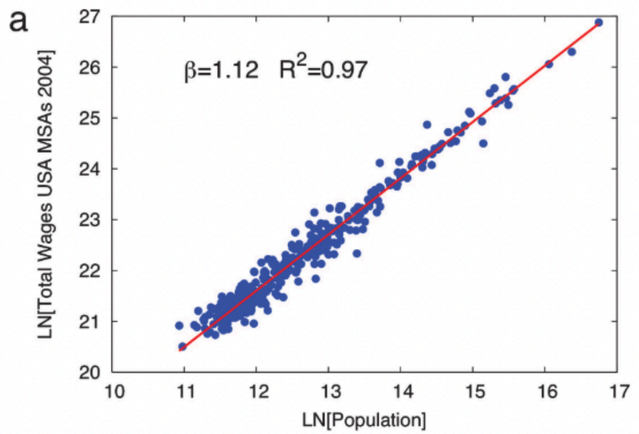



**Table 1. Scaling exponents for urban indicators vs. city size**

| *Y* | $\beta$ | 95% CI | Adj-$R^2$ | Observations | Country–year |
|---|---|---|---|---|---|
| New patents | 1.27 | [1.25,1.29] | 0.72 | 331 | U.S. 2001 |
| Inventors | 1.25 | [1.22,1.27] | 0.76 | 331 | U.S. 2001 |
| Private R&D employment | 1.34 | [1.29,1.39] | 0.92 | 266 | U.S. 2002 |
| "Supercreative" employment | 1.15 | [1.11,1.18] | 0.89 | 287 | U.S. 2003 |
| R&D establishments | 1.19 | [1.14,1.22] | 0.77 | 287 | U.S. 1997 |
| R&D employment | 1.26 | [1.18,1.43] | 0.93 | 295 | China 2002 |
| Total wages | 1.12 | [1.09,1.13] | 0.96 | 361 | U.S. 2002 |
| Total bank deposits | 1.08 | [1.03,1.11] | 0.91 | 267 | U.S. 1996 |
| GDP | 1.15 | [1.06,1.23] | 0.96 | 295 | China 2002 |
| GDP | 1.26 | [1.09,1.46] | 0.64 | 196 | EU 1999–2003 |
| GDP | 1.13 | [1.03,1.23] | 0.94 | 37 | Germany 2003 |
| Total electrical consumption | 1.07 | [1.03,1.11] | 0.88 | 392 | Germany 2002 |
| New AIDS cases | 1.23 | [1.18,1.29] | 0.76 | 93 | U.S. 2002–2003 |
| Serious crimes | 1.16 | [1.11, 1.18] | 0.89 | 287 | U.S. 2003 |
| Total housing | 1.00 | [0.99,1.01] | 0.99 | 316 | U.S. 1990 |
| Total employment | 1.01 | [0.99,1.02] | 0.98 | 331 | U.S. 2001 |
| Household electrical consumption | 1.00 | [0.94,1.06] | 0.88 | 377 | Germany 2002 |
| Household electrical consumption | 1.05 | [0.89,1.22] | 0.91 | 295 | China 2002 |
| Household water consumption | 1.01 | [0.89,1.11] | 0.96 | 295 | China 2002 |
| Gasoline stations | 0.77 | [0.74,0.81] | 0.93 | 318 | U.S. 2001 |
| Gasoline sales | 0.79 | [0.73,0.80] | 0.94 | 318 | U.S. 2001 |
| Length of electrical cables | 0.87 | [0.82,0.92] | 0.75 | 380 | Germany 2002 |
| Road surface | 0.83 | [0.74,0.92] | 0.87 | 29 | Germany 2002 |

Data sources are shown in *SI Text*. CI, confidence interval; Adj-$R^2$, adjusted $R^2$; GDP, gross domestic product.

Bettencourt, L. M. A., Lobo, J., Helbing, D., Kühnert, C., & West, G. B. (2007). Growth, innovation, scaling, and the pace of life in cities. *Proceedings of the National Academy of Sciences of the United States of America*, *104*(17), 7301–7306. https://doi.org/10.1073/pnas.0610172104

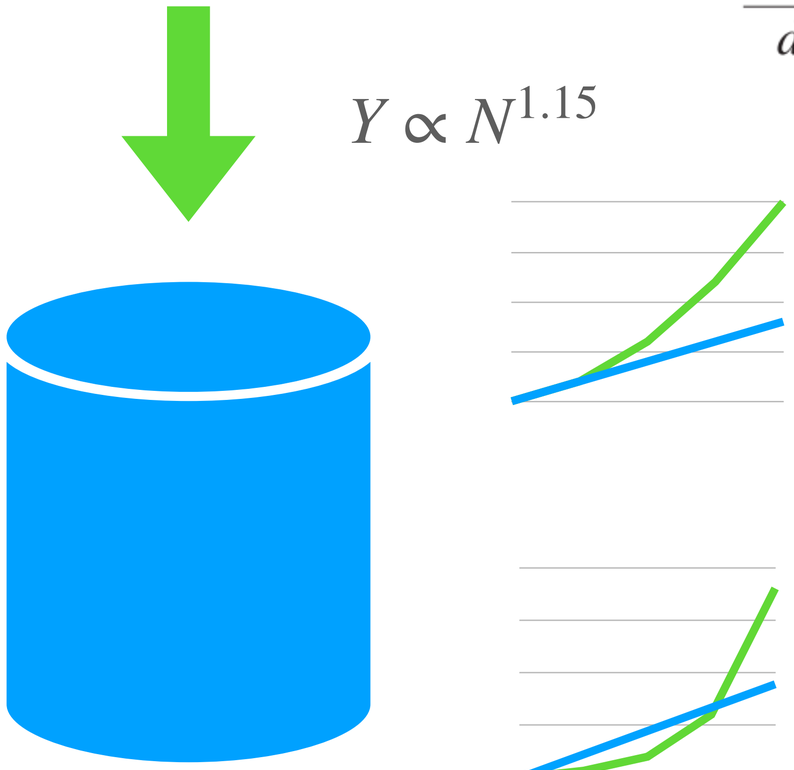
# Company Growth Modeling Based on Scaling Theory

## Scaling theory in City

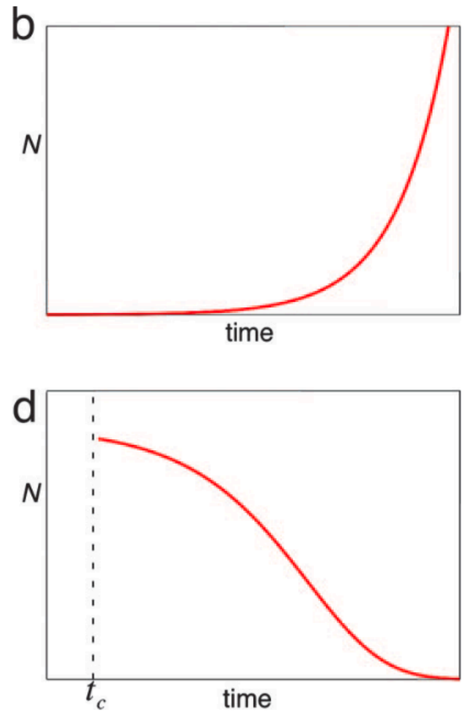
$$\frac{dN(t)}{dt} = \left(\frac{Y_0}{E}\right)N(t)^{\beta} - \left(\frac{R}{E}\right)N(t).$$

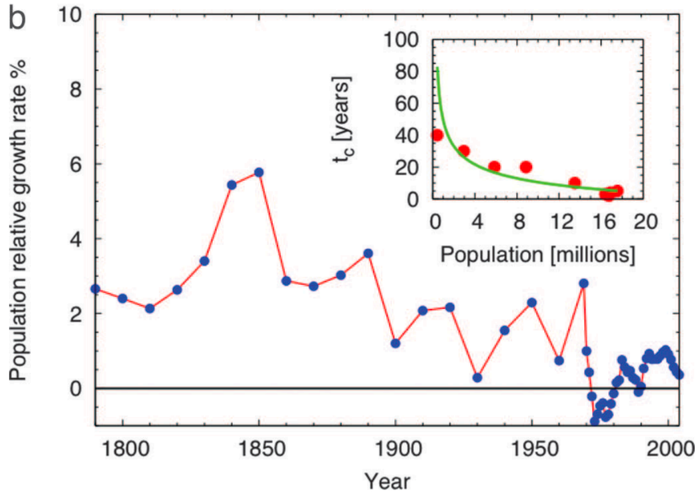
$$N(t) = \left[\frac{Y_0}{R} + \left(N^{1-\beta}(0) - \frac{Y_0}{R}\right)\exp\left[-\frac{R}{E}(1-\beta)t\right]\right]^{\frac{1}{1-\beta}}.$$

$Y \propto N^{1.15}$



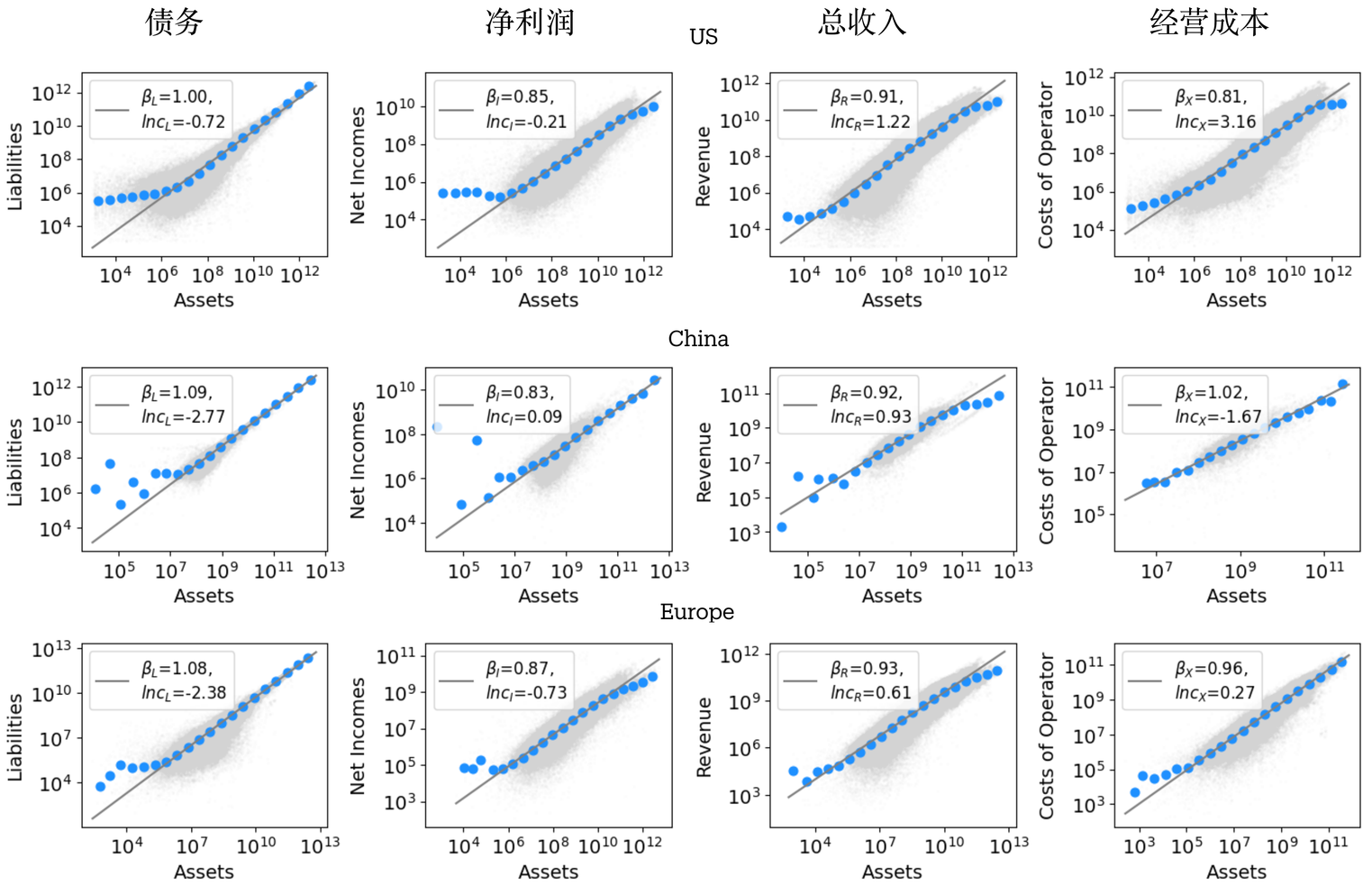
$Y \propto N$




Bettencourt, L. M. A., Lobo, J., Helbing, D., Kühnert, C., & West, G. B. (2007). Growth, innovation, scaling, and the pace of life in cities. *Proceedings of the National Academy of Sciences of the United States of America*, *104*(17), 7301–7306. https://doi.org/10.1073/pnas.0610172104

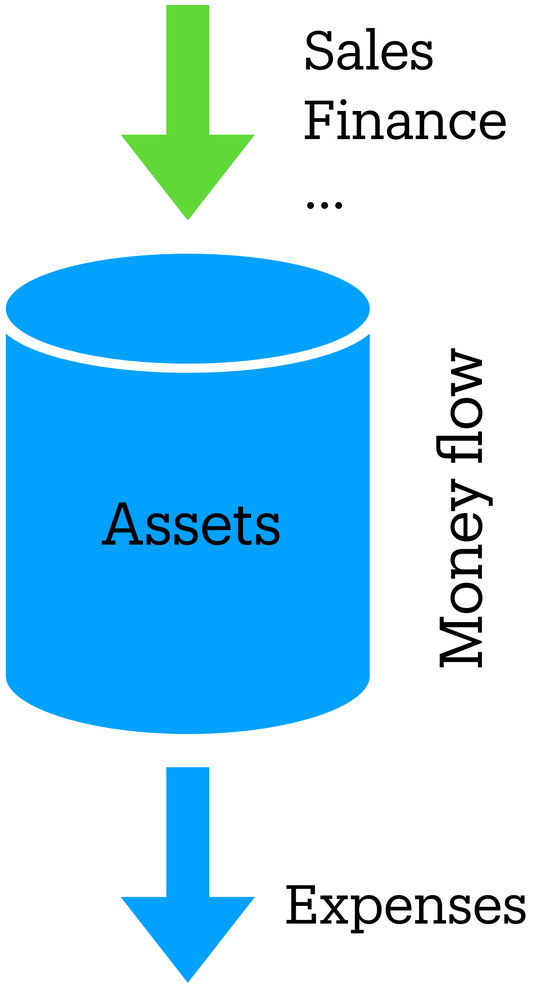
# Company Growth Modeling Based on Scaling Theory




Scale invariance has also been observed in company data, and it exhibits universality across different markets.

# Company Growth Modeling Based on Scaling Theory

## Establish a scientific company growth model



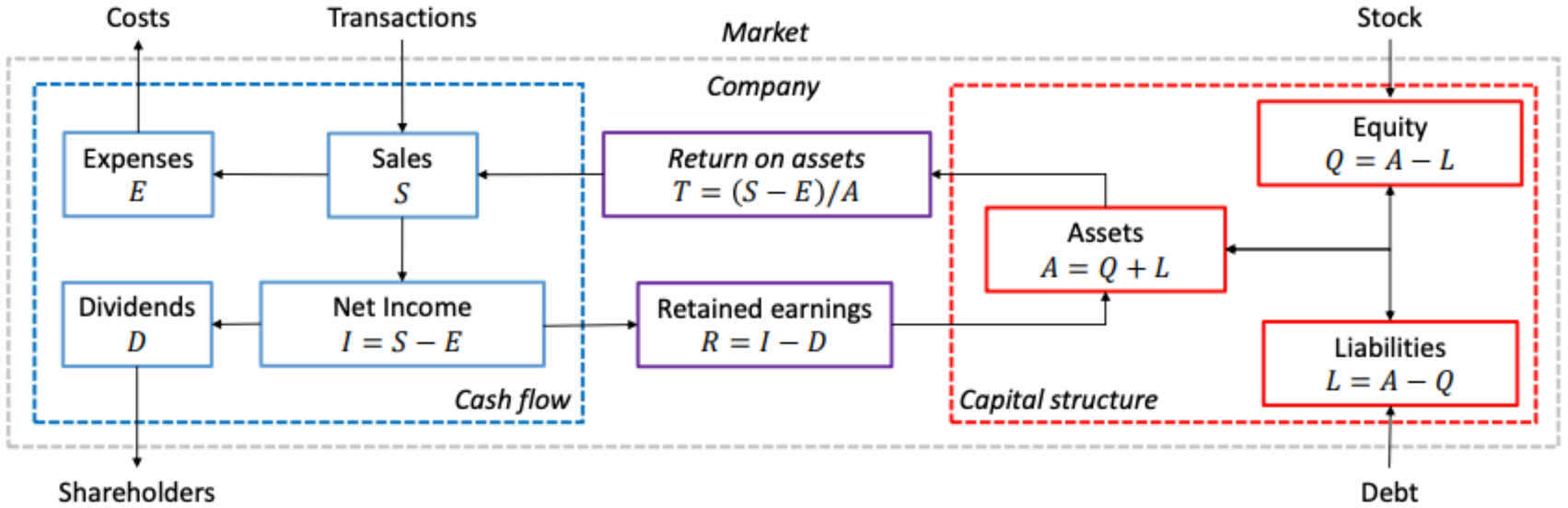

$$\Delta A(t) = [S(t) + F(t) - E(t)]\Delta t$$

$\Delta t \to 0$

$dL \approx F$

$$\frac{dA}{dt} = I + \frac{dL}{dt}$$

$$I = c_I A^{\beta_I}$$
$$L = c_L A^{\beta_L}$$

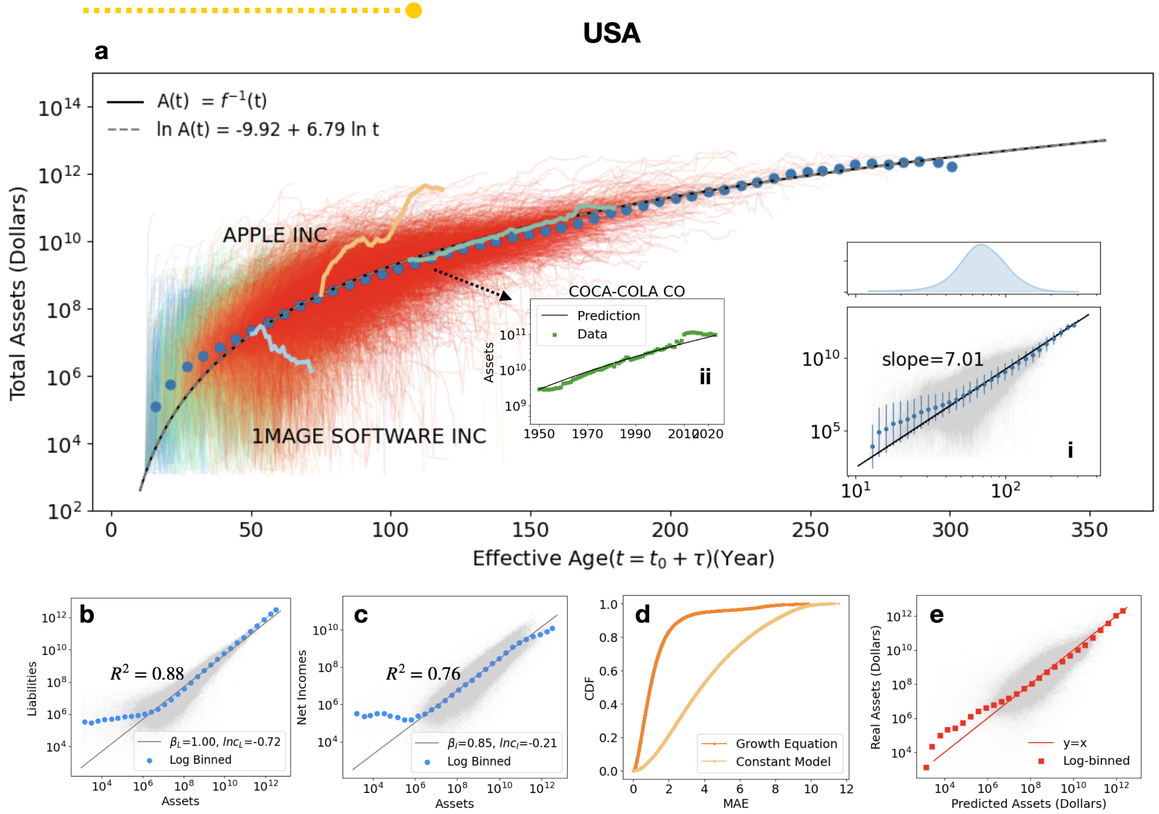


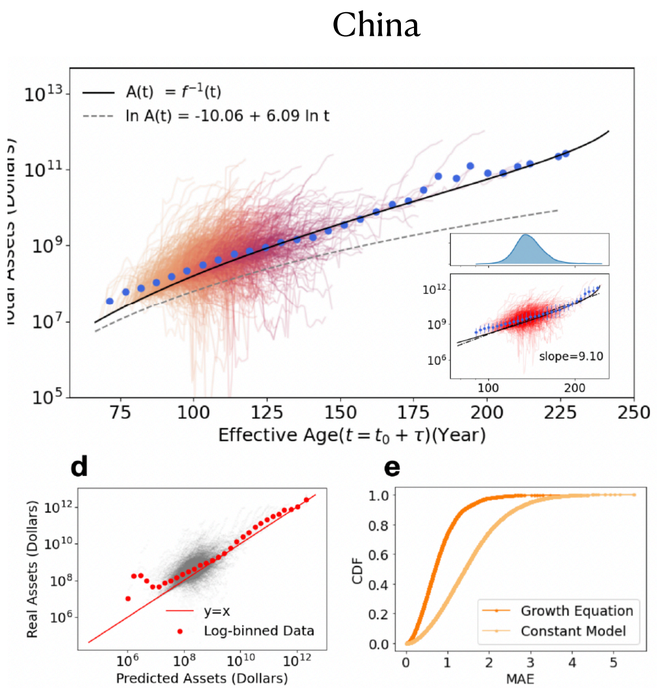

$$\frac{dA}{dt} = \frac{c_I A^{\beta_I}}{1 - c_L \beta_L A^{\beta_L - 1}}$$

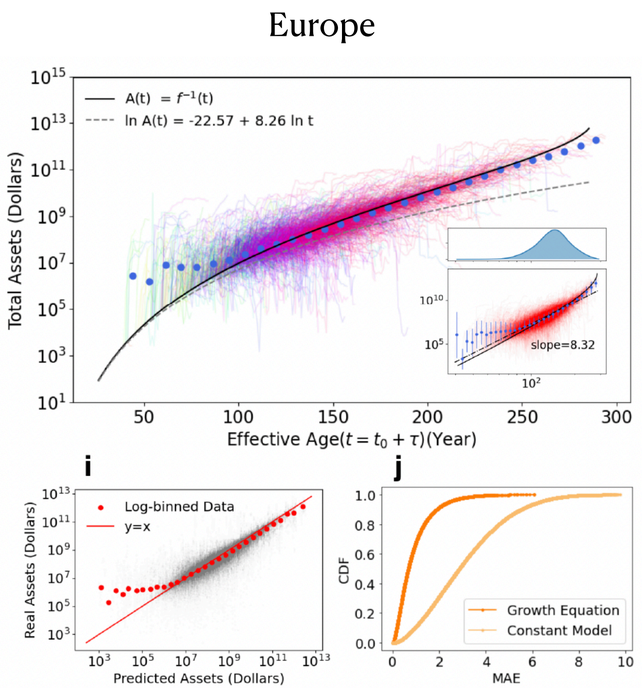

**Tao R &** Zhang J, et al. Scaling laws and a general theory for the growth of public companies. arXiv preprint arXiv:2109.10379, 2024.

# Company Growth Modeling Based on Scaling Theory

## Establish a scientific company growth model



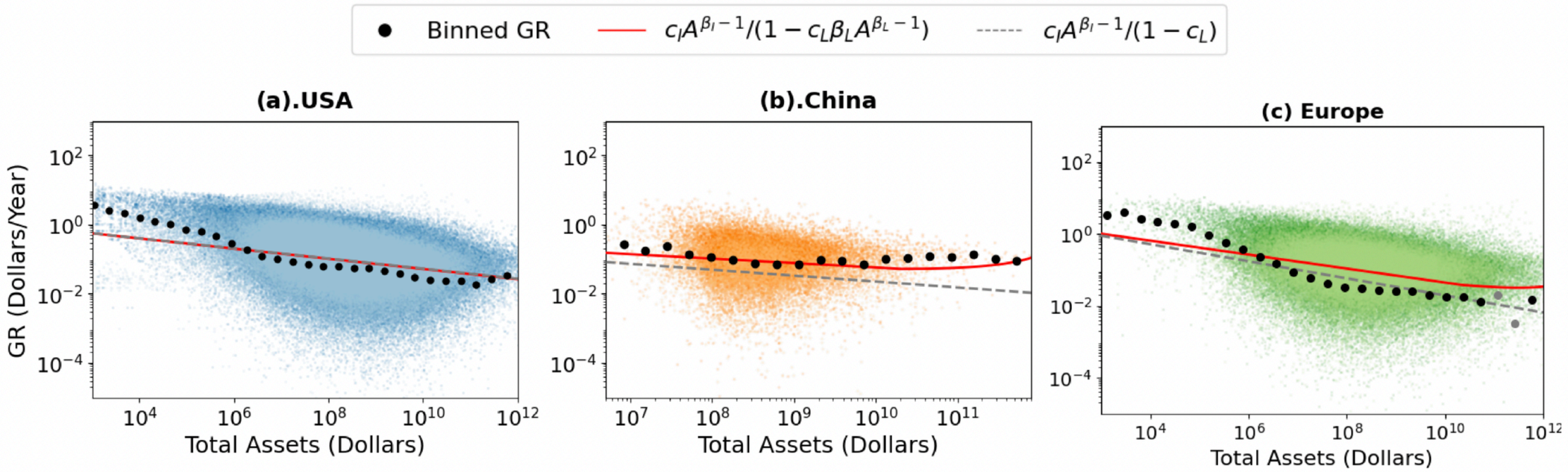




- The model has been validated using data of listed companies in three markets: the United States, China, and Europe.


**Tao R &** Zhang J, et al. Scaling laws and a general theory for the growth of public companies. arXiv preprint arXiv:2109.10379, 2024.

# Company Growth Modeling Based on Scaling Theory

## Establish a scientific company growth model




- Previous studies concluded that the growth rate of small companies is negatively correlated with size, while the growth of large enter companies is independent of size. Our model reconciles the debates over the relationship between growth rate and size.


**Tao R &** Zhang J, et al. Scaling laws and a general theory for the growth of public companies. arXiv preprint arXiv:2109.10379, 2024.

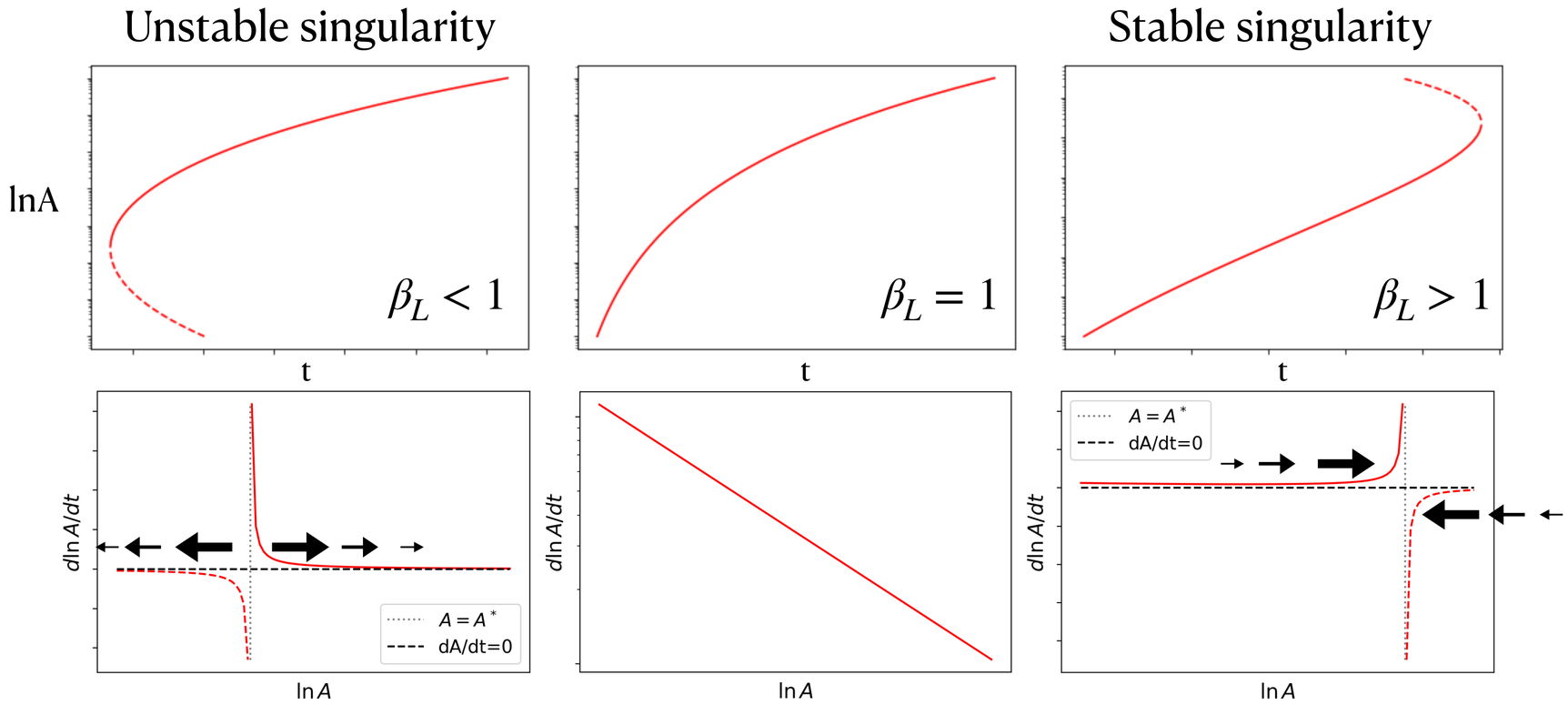
# Company Growth Modeling Based on Scaling Theory

## Upper limit and entry threshold for company growth

$$\frac{dA}{dt} = \frac{c_I A^{\beta_I}}{1 - c_L \beta_L A^{\beta_L - 1}}$$

$$\frac{dt}{dA} = 0 \implies A^* = c_L \beta_L^{1/(1-\beta_L)}$$

$$\left.\frac{d^2 t}{dA^2}\right|_{A=A^*} = \frac{(A^*)^{-\beta_I - 1}}{c_I}(1 - \beta_L)$$




- Dynamics is most sensitive to changes in the $\beta_L$ (LT vs AT)
- The model predicts the existence of the company growth upper limit ($\beta_L > 1$) and the growth threshold ($\beta_L < 1$)

# Company Growth Modeling Based on Scaling Theory

## Summary


- Scaling law indictors the scale invariance and complexity for system.
- Scaling Theory builds the bridge of scaling law to system growth
- We proposed a company growth equation based on Scaling theory, which is consistent with the data, and reveals the constraints on company growth.

北京师范大学系统科学学院
BEI JING SHI FAN DA XUE XI TONG KE XUE XUE YUAN

# Thanks